2019 EXAMINATIONS

Lancaster University

Part II

COMPUTING AND COMMUNICATIONS [2.5 Hours]

SCC.366 Media Coding and Processes

Candidates are asked to answer THREE questions from FOUR; each question is worth a total of 25 marks.

Use a separate answer book for each question.

Simple calculators permitted

Question 1

This question covers basic **image and audio coding** concepts.

1.a

i. Explain in detail how an audio signal is encoded and decoded using PCM and ADPCM and compare their coding efficiency.

[2.5 marks]

ii. Describe different types of masking effects found in an audio signal and explain how masking is used to improve the level of compression.

[2.5 marks]

1.b MPEG-4 defines coding types for Rectangular Video and Arbitrarily Shaped Video.

i. Explain the difference between these two concepts. What is the idea behind Arbitrarily Shaped Video and what concepts have been standardised to achieve this? **[3 marks]**

ii. Describe how a scene can be created in the case of Arbitrarily Shaped Video.

[2 marks]

1.c Describe layered coding of video signals. **[5 marks]**

1.d The figure below shows the frequency spectrum of a signal

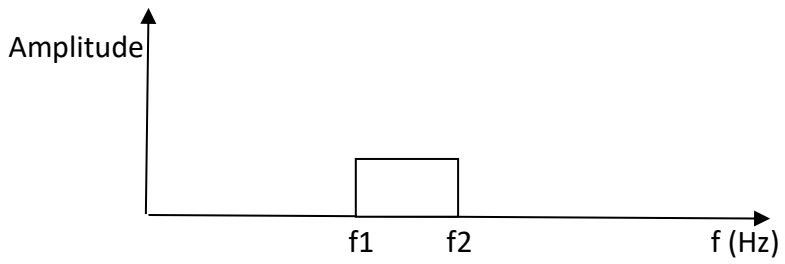

Figure 1

i. Determine the minimum required sampling rate for this signal so it can be reconstructed form its sampled version? Provide an explanation for this.

[5 marks]

ii. If f2 = 1200 KHz, and f1= 1120 KHz what is the minimum required bit rate if the samples are quantised using a quantisation system with 1024 equal quantization steps.

[5 marks]

Total 25 marks

Question 2

This question covers **Data Coding** concepts.

2.a Consider a source symbol generation process where the source symbols are given as a set with 7 elements: S = {A,B,C,D,E,F,G}, and the probability of the occurrence of each symbol in S is given as: P(A)= P(B)=P(C)=P(D)=P(E)=P(F)=1/7. Design Huffman code for this source.

[8 marks]

2.b With two examples show that for a set of symbols with equal probabilities the Huffman code has always a compression ratio smaller than 1. Provide an explanation for this.

[7 marks]

2.c Provide the results of run-length encoding for the two source sequences that are given below. The outputs are given as sequences of symbols in S={A,B,C,D,E,F} and natural numbers.

[BBCCCCCCCCCCBBBBBAAAAADDDDDD]

[BCCCBCCBCCBCBBACCABADDDAADDD]

[5 marks]

2.d Describe the main difference between coding for error detection, and coding for compression.

[5 marks]

Total 25 marks

Question 3

This question is about the **MPEG video standard** and **Video Processing**

3.a In MPEG coding

i. Describe I, B, and P frames in the MPEG standard and explain how they shape a group of pictures (GOP).

[5 marks]

ii. Consider two GOP with the same number of frames. GOP1 consists of I and P frames, and GOP2 consists of I and B frames. Which one has a better video quality? Which one has a higher number of bits? Explain your answers.

[3 marks]

iii. In an MPEG coder, describe the function of Motion Estimation Block and explain what kind of redundancy is exploited by this block.

[2 marks]

3. b This part of the question is about **edge and cut detection**.

i. Describe the region growing strategy used to deal with detecting edges in a noisy image and formulate a pseudo code algorithm for this method.

[7 marks]

3.c This part of the question is about **line detection**. We assume that a black-and-white version of the source image (spatial space) is available, where pixels that are located on an edge are set to black.

i. Describe an algorithm to detect straight lines in spatial domain in an arbitrary source image using the Hough transform.

[5 marks]

Question 3 continues on next page…

Question 3 continued.

ii. Figure 3.1 shows an image. Explain how the curves in Hough space help in detecting the lines in the image.

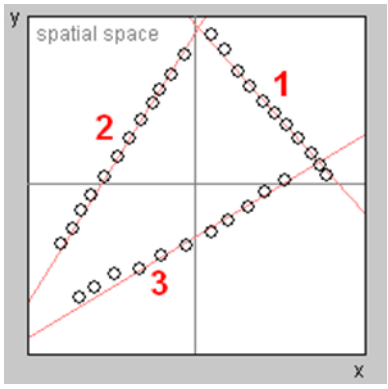

Fig3.1

[3 marks]

Total 25 marks

Question 4

This question is about video **watermarking**.

4.a

i. Explain what a Digital Watermark consists of in an image and describe 3 characteristics of an invisible watermark.

[5 marks]

ii. Describe the possible attacks on a digital watermark (at least two type of attacks).

[4 marks]

iii. Describe format specific watermarking and explain its advantages and disadvantages.

[8 marks]

4.b Assume that C1 and C2 are codewords where C1.C2=0 and C1.C1=C2.C2=1. Design a watermarking system based on C1 and C2 and show its block diagram. Check whether such a watermarking system fulfils the desired characteristic of an invisible watermark.

[8 marks]

Total 25 marks

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